

Sharp Weighted Type Under Formal Reversion

Zero-loss inversion, exact Borel radii, and
non-D-finiteness in countable exponential feedback

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29 September 2026

Abstract

Let $M_0 = 1$ and let (M_n) be a positive log-convex sequence. We prove that the weighted coefficient type

$$\tau_M(f) = \limsup_{n \rightarrow \infty} \left(\frac{|[z^n]f|}{M_n} \right)^{1/n}$$

is preserved by every tangent-to-identity formal compositional inverse if and only if $M_n^{1/n} \rightarrow \infty$. A finite Lagrange majorant gives an explicit overhead $\exp\{O(n/\sqrt{M_n^{1/n}})\}$, hence no loss in the exponential constant. Analytic coordinate changes obey an exact scaling law, without a moderate-growth assumption on the weight.

Using two explicitly stated forward estimates from ProveIt's countable exponential-feedback report, we obtain the missing sharp type of its inverse, exact inverse Borel radii, and a logarithmically refined inverse type. For $\lambda_j = j^2$, this yields

$$\limsup_{n \rightarrow \infty} \left(\frac{|[u^n]\widehat{Q}(u)| \log(n+e)^{2n}}{n!} \right)^{1/n} = 4.$$

Its ordinary Borel transform is entire but has infinite classical order, with an exact iterated-logarithmic maximum-modulus limsup. An elementary holonomic obstruction proves that both the forward series and its inverse are non-D-finite. All new arguments are proved in the article; the two repository inputs, finite computational checks, and remaining research questions are distinguished explicitly.

Status. Conventional mathematical proofs and reproducible exact finite checks; not Lean-verified or peer-reviewed. The theorem is a proposed contribution and resolves a documented repository-level gap. Global publication priority, and a solution of a named longstanding conjecture, are not claimed.

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Reading guide. Sections 2–6 form the independent core: a finite coefficient estimate, exact inversion invariance, its necessary-and-sufficient weight condition, and analytic coordinate covariance. Section 7 states the two repository inputs and derives the feedback applications. Sections 8 and 9 convert the refined coefficient type into entire growth and a non-D-finiteness obstruction. Finite verification and the proposed research program are kept separate in Sections 10 and 11.

1 Research target, contribution, and dependency boundaries

1.1 The precise gap being addressed

The current ProveIt development separates its transseries library from the Fabius-function library. The relevant research directory is now

`Analysis/Transseries/docs/series-and-transseries/`.

The repository was inspected at commit `0c973d8f5e7ef4550f4df1bf486d890037fd9f14`. Its transseries README separates the existing Lean development from the accompanying research articles [1, 2]. This article continues the package *A sharp regularity classification for countable exponential-feedback transseries* [3], rather than treating another special-function inverse by a finite coefficient calculation.

That package studies

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q)), \quad \lambda_j \geq 0, \quad \widehat{Q} = \widehat{U}^{-1}, \quad (1.1)$$

where the inverse is compositional. It gives an exact forward Gevrey type and logarithmic forward coefficient asymptotics, but its README explicitly limits the reversion theorem to class membership, not the sharp numerical type. This is a substantive distinction. Knowing that coefficients are bounded by $CA^n(n!)^s$ for some A does not determine the smallest possible exponential rate, nor the exact radius of the factorial-normalized Borel series.

The gap is resolved here by a theorem that is independent of the feedback model. We preserve the *exact limsup type* under formal inversion for a large class of coefficient weights, determine precisely when this universal preservation property holds within the log-convex class, and apply it to (1.1). The forward coefficients are positive, while inverse coefficients can have cancellations. The theorem shows that cancellation cannot lower their weighted exponential limsup.

1.2 Why a general weight theorem is useful

A factorial scale alone loses the information relevant to the quadratic model. Its series have Gevrey order one but zero ordinary Gevrey-one type. The finer scale

$$M_n = \frac{n!}{\log(n + e)^{2n}}$$

has nonzero finite type. Preservation on this scale is substantially more informative than preservation of Gevrey membership. The same proof applies to weights growing much more slowly than every positive factorial power, and also to very rapidly growing weights for which no moderate-growth property is available.

The proof uses only finite coefficient algebra, log-convexity, and a subexponential estimate. It does not require an analytic solution of the feedback equation, a summation operator, positivity of inverse coefficients, or a description of their signs. This makes the result suitable as an independent formalization target.

1.3 What is proved here and what is imported

All results in Sections 2–6 are proved from elementary formal power-series identities. In Section 7, two forward estimates are restated as *Repository inputs*: the exact forward Gevrey type and the regularly varying forward coefficient asymptotic. Those two results are used as mathematical inputs from [3]; their full proofs are not independently re-audited in this article. The inverse

consequences and the growth and non-D-finiteness arguments are proved here from those stated inputs.

The status of each family of assertions is summarized in Table 1. In particular, no application is presented as Lean-verified merely because it continues a repository that also contains Lean code.

Result	Dependency and status
Weight concentration; finite inverse majorant	Proved here by finite inequalities and formal Lagrange inversion.
Exact type and sharp threshold	Proved here from the finite majorant; no feedback theorem used.
Analytic coordinate covariance	Proved here from concentration and the inverse theorem.
Forward feedback types and asymptotics	Two explicitly restated results from the conventional research report [3].
Inverse feedback types and exact Borel radii	New deductions here from those inputs and the general inverse theorem.
Entire growth and non-D-finiteness	Elementary proofs here; application to feedback uses its imported forward asymptotic.
Computational evidence	Exact checks at finite orders; neither asymptotic proof nor formal verification.

Table 1: Dependency and verification ledger.

1.4 Relationship to the literature

Lagrange inversion is classical; Gessel’s survey [5] describes its formal forms and proofs. Stability of ultradifferentiable classes under composition and inversion is an established subject, including the weight-sequence and weight-matrix results of Rainer and Schindl [6, 7]. The present object is the exact coefficient limsup of a *single formal germ*; no realization as an ultradifferentiable function is presumed. This distinction specifies our statement, but is not a claim that the literature contains no related or stronger formulation.

Borel convergence, Borel continuation, and Borel–Laplace summability are different properties; see Sauzin [8]. We maintain that distinction throughout. The negative-ray package already in ProveIt [4] addresses fine negative-direction summation. The present article does not replace that result with a maximum-modulus assertion or claim angular summability from a coefficient bound.

The source review was targeted: the current directory inventory, the transseries README, the classification report’s stated theorems and scope, and the negative-ray report’s summary were consulted. It was not a claim-by-claim audit of the full canonical volume or a complete publication priority search. The general theorem and its applications are proposed for independent mathematical review.

Editorial note (ProveIt, 2026-09-29). (i) The gap closed here is also posed in [3] itself, as its research question “Exact type under compositional inversion” (`Exponential_Feedback_Regularity_Classification/article.tex`, lines 1544–1549); this article cites only the corresponding README sentence. (ii) The inspected snapshot already contained the finite-core package [9] (filed in batch 47), which this article does not cite. For $\lambda_j = aj^p$ with $p > 2$ its Theorem `thm:explicit` proves $v_n \sim -u_n$ for the inverse coefficients, which already implies the inverse type of Theorem 7.6 in that case ($\beta = 0$). The results here are new for $1 < p \leq 2$, for $\beta \neq 0$, and for nonnegative slopes without regular variation. The question “Full inverse asymptotics for quadratic feedback” in Section 11 re-poses that package’s Conjecture `conj:quadratic-inverse`; see the note there. (iii) For $\beta = 0$ the forward half of Theorem 8.2 is Theorem `thm:borelintro` of [3]; see the note

after it. (iv) “Sharp” and “exact” in the title and abstract refer to limsup (n th-root) types, radii of convergence and maximum-modulus limsups. Limits, signs and subexponential factors of the inverse coefficients are not claimed here.

2 Log-convex weights and coefficient type

2.1 Definitions and elementary type rules

All formal power series are over $\mathbb{C}$, unless otherwise stated. A formal diffeomorphism is a series $f(z) = az + O(z^2)$ with $a \neq 0$. Such a series has a unique compositional inverse. A *weight* is a positive sequence $M = (M_n)_{n \geq 0}$ with $M_0 = 1$.

Definition 2.1. A weight is log-convex if

$$M_n^2 \leq M_{n-1}M_{n+1} \quad (n \geq 1). \quad (2.1)$$

For $n \geq 1$, write $r_n = M_n^{1/n}$. The weight is called superexponential when $r_n \rightarrow \infty$. For $f = \sum_{n \geq 0} a_n z^n$, set

$$\tau_M(f) = \limsup_{n \rightarrow \infty} \left(\frac{|a_n|}{M_n} \right)^{1/n} \in [0, \infty]. \quad (2.2)$$

The choice of M_n refers to *coefficients*, not derivatives. Thus $M_n = (n!)^s$ gives the factorial-normalized Gevrey type used in this article. All equalities involving types permit zero and infinity. Positive finite scalar multiplication of an extended type is interpreted in the usual way.

Lemma 2.2 (Elementary type calculus). *Finite changes of coefficients or weights do not affect (2.2). For $c \neq 0$,*

$$\tau_M(cf) = \tau_M(f), \quad \tau_M(f(cz)) = |c|\tau_M(f).$$

Furthermore,

$$\tau_M(f + h) \leq \max\{\tau_M(f), \tau_M(h)\}.$$

If the two types on the right are unequal, equality holds. If M is superexponential, every convergent power series has type zero, and adding any convergent power series preserves the type of an arbitrary formal series.

Proof. The first statements follow directly by taking n th roots. For the sum, $|a_n + b_n| \leq 2 \max(|a_n|, |b_n|)$ and $2^{1/n} \rightarrow 1$. If the types are unequal, write $f = (f + h) - h$ to rule out strict inequality below the larger one. A convergent series has coefficients bounded by CR^{-n} , so its normalized n th root is at most $C^{1/n}R^{-1}/r_n \rightarrow 0$. When both types are zero, the sum also has type zero; this completes the last assertion. $\square$

2.2 The two concentration inequalities

Lemma 2.3 (Log-convex concentration). *Let M be log-convex. Then $\mu_n = M_n/M_{n-1}$ and r_n are nondecreasing. For $0 \leq \ell \leq n$,*

$$\frac{M_{n-\ell}}{M_n} \leq r_n^{-\ell}. \quad (2.3)$$

For positive integers $m_1, \dots, m_k$ with sum $n - 1$,

$$\prod_{i=1}^k M_{m_i+1} \leq M_{n-k+1} M_2^{k-1}. \quad (2.4)$$

For positive integers $j_1, \dots, j_k$ with sum n ,

$$\prod_{i=1}^k M_{j_i} \leq M_{n-k+1} M_1^{k-1}. \quad (2.5)$$

Proof. Log-convexity is equivalent to monotonicity of μ_n . Since $\log r_n$ is the average of the first n terms of the nondecreasing sequence $\log \mu_n$, it is nondecreasing. The polygonal function with values $\log M_n$ at the integers is convex and is zero at the origin. Its value at $n - \ell$ is therefore at most $(n - \ell) \log M_n / n$. Exponentiating proves (2.3).

For (2.4), consider the indices $m_i + 1 \geq 2$. If two indices $a \geq b$ satisfy $b > 2$, transfer one unit from b to a . The product cannot decrease, because

$$\frac{M_{a+1} M_{b-1}}{M_a M_b} = \frac{\mu_{a+1}}{\mu_b} \geq 1.$$

Repeat until all but one index equal two. The remaining index is $n - k + 1$, since the original total is $n - 1 + k$. This proves (2.4). The same argument, with lower bound one in place of two, proves (2.5). $\square$

These inequalities do not require $M_1 \geq 1$ or monotonicity of the weights themselves. Only positivity and log-convexity are used. The factor $r_n^{-(k-1)}$ obtained from (2.3) is the mechanism that suppresses the combinatorial multiplicity of nonlinear inverse terms.

3 A finite coefficient bound for formal reversion

3.1 The coefficient formula

Write

$$f(z) = z + \sum_{j \geq 2} a_j z^j = z(1 + h(z)), \quad g(z) = f^{-1}(z) = z + \sum_{n \geq 2} b_n z^n. \quad (3.1)$$

Lagrange inversion gives, for $n \geq 1$,

$$b_n = \frac{1}{n} [z^{n-1}] (1 + h(z))^{-n}. \quad (3.2)$$

A formal residue proof is included in Appendix A; no analytic convergence is required. Expanding the negative integer power in (3.2), for $n \geq 2$ one obtains the finite identity

$$b_n = \frac{1}{n} \sum_{k=1}^{n-1} (-1)^k \binom{n+k-1}{k} \sum_{\substack{m_1 + \dots + m_k = n-1 \\ m_i \geq 1}} \prod_{i=1}^k a_{m_i+1}. \quad (3.3)$$

The term $k = 1$ equals $-a_n$. The other terms can be large and can cancel; it would be invalid to infer a lower bound for $|b_n|$ by keeping only that first term. Our proof instead obtains a one-sided upper bound that can be applied twice, once in each direction of inversion.

Theorem 3.1 (Finite inverse majorant). *Assume M is log-convex and*

$$|a_j| \leq C A^j M_j \quad (j \geq 2), \quad C, A > 0. \quad (3.4)$$

Set $D = CA$. For every $n \geq 2$,

$$|b_n| \leq \frac{A^{n-1}}{n} \sum_{k=1}^{n-1} \binom{n+k-1}{k} \binom{n-2}{k-1} D^k M_{n-k+1} M_2^{k-1}. \quad (3.5)$$

In particular, with

$$X_n = \frac{2DM_2n^2}{r_n}, \quad (3.6)$$

one has the explicit bound

$$\begin{aligned} |b_n| &\leq 2DA^{n-1}M_n \sum_{\ell=0}^{n-2} \frac{X_n^\ell}{\ell!(\ell+1)!} \\ &\leq 2DA^{n-1}M_n \exp\left(2n\sqrt{\frac{2DM_2}{r_n}}\right). \end{aligned} \quad (3.7)$$

Proof. The number of positive ordered compositions of $n-1$ into k parts is $\binom{n-2}{k-1}$. Each summand in the inner sum of (3.3) is bounded by

$$C^k A^{n-1+k} \prod_i M_{m_i+1} \leq A^{n-1} D^k M_{n-k+1} M_2^{k-1},$$

by Lemma 2.3. Taking absolute values proves (3.5).

For $1 \leq k \leq n-1$, use

$$\binom{n+k-1}{k} \leq \frac{(2n)^k}{k!}, \quad \binom{n-2}{k-1} \leq \frac{n^{k-1}}{(k-1)!}, \quad \frac{M_{n-k+1}}{M_n} \leq r_n^{-(k-1)}.$$

Substitution in (3.5), followed by $\ell = k-1$, gives the first inequality in (3.7). For $X \geq 0$,

$$\begin{aligned} \sum_{\ell \geq 0} \frac{X^\ell}{\ell!(\ell+1)!} &\leq \sum_{\ell \geq 0} \frac{X^\ell}{(\ell!)^2} \\ &= \sum_{\ell \geq 0} \binom{2\ell}{\ell} \frac{X^\ell}{(2\ell)!} \leq \sum_{\ell \geq 0} \frac{(2\sqrt{X})^{2\ell}}{(2\ell)!} \leq \exp(2\sqrt{X}). \end{aligned}$$

This proves the second inequality. $\square$

3.2 An effective zero-loss interpretation

The finite sum (3.5) is an exact-arithmetic bound when the data are rational. The exponential simplification is more convenient for asymptotics. In particular,

$$\left(\frac{|b_n|}{M_n}\right)^{1/n} \leq A \left(\frac{2D}{A}\right)^{1/n} \exp\left(2\sqrt{\frac{2DM_2}{r_n}}\right). \quad (3.8)$$

Every factor beyond A tends to one if $r_n \rightarrow \infty$.

For a concrete tolerance $\varepsilon > 0$, the condition

$$r_n \geq \frac{8DM_2}{\log(1+\varepsilon)^2}$$

makes the exponential factor in (3.8) at most $1+\varepsilon$. A separate finite lower bound on n makes $(2D/A)^{1/n} \leq 1+\varepsilon$ as well. Thus the theorem gives a computable onset for an inverse rate $A(1+\varepsilon)^2$, provided a verified input bound and growth bound for r_n are supplied. A mere limsup statement need not supply those bounds effectively.

4 Exact type invariance and its sharp boundary

Theorem 4.1 (Zero-loss reversion). *Let M be positive and log-convex, with $M_0 = 1$ and $r_n \rightarrow \infty$. For every formal diffeomorphism $f(z) = az + O(z^2)$,*

$$\tau_M(f^{-1}) = \frac{\tau_M(f)}{|a|}. \quad (4.1)$$

In particular, tangent-to-identity inversion preserves type exactly, including zero and infinite type.

Proof. First suppose $a = 1$ and $\tau_M(f) < \infty$. For every $A > \tau_M(f)$, definition (2.2) gives a constant $C > 0$ such that (3.4) holds at every $j \geq 2$: finitely many initial coefficients are absorbed in C . Theorem 3.1 and (3.8) give $\tau_M(f^{-1}) \leq A$. Letting A decrease to $\tau_M(f)$ proves

$$\tau_M(f^{-1}) \leq \tau_M(f).$$

This includes $\tau_M(f) = 0$, since every $A > 0$ may be used. The inverse also has finite type, so apply the same argument to f^{-1} , whose inverse is f , to obtain the reverse inequality.

If $\tau_M(f) = \infty$ but $\tau_M(f^{-1}) < \infty$, the already proved finite-type upper bound applied to f^{-1} would imply $\tau_M(f) < \infty$, a contradiction. This handles the infinite case.

For arbitrary $a \neq 0$, set $F = f/a$. Output scaling does not change type and $F'(0) = 1$. Since

$$F^{-1}(z) = f^{-1}(az),$$

Lemma 2.2 and the tangent case give $|a|\tau_M(f^{-1}) = \tau_M(F^{-1}) = \tau_M(F) = \tau_M(f)$. $\square$

Theorem 4.2 (Necessary and sufficient weight condition). *For a positive log-convex weight $M_0 = 1$, the following are equivalent:*

- (i) $M_n^{1/n} \rightarrow \infty$.
- (ii) Every tangent-to-identity formal diffeomorphism satisfies $\tau_M(f^{-1}) = \tau_M(f)$.
- (iii) The inverse of every tangent-to-identity polynomial has type zero.

Proof. The implication (i) $\Rightarrow$ (ii) is Theorem 4.1, and (ii) $\Rightarrow$ (iii) follows because polynomials have type zero. For the converse, the roots r_n are positive and nondecreasing, so failure of (i) means $r_n \rightarrow L$ for some $0 < L < \infty$. Take $f(z) = z - z^2$. Its inverse is

$$f^{-1}(z) = \frac{1 - \sqrt{1 - 4z}}{2} = \sum_{n \geq 1} \frac{1}{n} \binom{2n-2}{n-1} z^n. \quad (4.2)$$

The coefficients have n th-root limit 4, by Stirling's formula. Consequently $\tau_M(f^{-1}) = 4/L > 0$, contradicting (iii). $\square$

Remark 4.3 (The analytic endpoint is genuinely different). For $M_n = 1$, type is the reciprocal of the ordinary radius of convergence. In (4.2), a polynomial of type zero has an inverse of type four. Thus no version of exact type preservation that includes all analytic radii can hold. Superexponential normalization makes every analytic germ type zero and removes this analytic singularity obstruction from the coefficient scale.

4.1 What exact type does and does not say

If $0 < \tau_M(f) = T < \infty$, the equality in Theorem 4.1 says that no uniform exponential improvement of the inverse coefficients is possible. For every $0 < \varepsilon < T$, infinitely many indices satisfy

$$|b_n| \geq (T - \varepsilon)^n M_n$$

in the tangent case, while for every $\varepsilon > 0$ all sufficiently large indices satisfy

$$|b_n| \leq (T + \varepsilon)^n M_n.$$

These statements are compatible with zeros, sign changes, sparse large coefficients, and substantial subexponential cancellation. They do not imply that $|b_n|^{1/n}/r_n$ has a limit. Reversion twice is what proves the sharp lower limsup; no unproved sign assumption enters.

5 Exact covariance under analytic coordinate changes

5.1 Postcomposition: a second finite estimate

Proposition 5.1 (Analytic outer functions). *Assume M satisfies the hypotheses of Theorem 4.1. Let $f(0) = 0$ be any formal series and let H be analytic near zero. Then*

$$\tau_M(H \circ f) \leq \tau_M(f). \quad (5.1)$$

If $H'(0) \neq 0$, equality holds. More explicitly, suppose

$$|[z^j]f| \leq CA^j M_j \quad (j \geq 1), \quad |[w^k]H| \leq BR^{-k} \quad (k \geq 1).$$

For $n \geq 1$,

$$|[z^n]H(f(z))| \leq \frac{BC}{R} A^n M_n \exp\left(\frac{CM_1 n}{Rr_n}\right). \quad (5.2)$$

Proof. There are $\binom{n-1}{k-1}$ positive compositions of n into k parts. Lemma 2.3 gives

$$\begin{aligned} |[z^n]H(f(z))| &\leq BA^n \sum_{k=1}^n (C/R)^k \binom{n-1}{k-1} M_{n-k+1} M_1^{k-1} \\ &\leq \frac{BC}{R} A^n M_n \left(1 + \frac{CM_1}{Rr_n}\right)^{n-1}, \end{aligned}$$

which implies (5.2). For finite $\tau_M(f)$, choose any $A > \tau_M(f)$ and absorb all initial coefficients in C . Taking n th roots and then the limsup proves (5.1); if the right side is infinite there is nothing to prove. If $H'(0) \neq 0$, the analytic inverse germ of $H - H(0)$ exists. Apply the upper inequality to this inverse and to $H(f) - H(0)$, then use invariance under finite coefficient changes. $\square$

5.2 Precomposition and conjugacy

Theorem 5.2 (Analytic coordinate covariance). *Let M satisfy the hypotheses of Theorem 4.1. Let f be any formal series, let g be analytic with $g(0) = 0$ and $g'(0) = c \neq 0$, and let H be analytic near $f(0)$ with $H'(f(0)) \neq 0$. Then*

$$\tau_M(H \circ f \circ g) = |c| \tau_M(f). \quad (5.3)$$

In particular, conjugacy by an analytic tangent-to-identity change of variable preserves exact type.

Proof. Start with a formal diffeomorphism f and write $a = f'(0) \neq 0$. The inverse factorization is

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

The germ g^{-1} is analytic and has nonzero derivative, so Proposition 5.1 and Theorem 4.1 give

$$\frac{\tau_M(f \circ g)}{|ac|} = \tau_M((f \circ g)^{-1}) = \tau_M(f^{-1}) = \frac{\tau_M(f)}{|a|}.$$

Thus $\tau_M(f \circ g) = |c|\tau_M(f)$.

For general f , subtract its constant term and choose a linear polynomial ℓ so that $F = f - f(0) + \ell$ has nonzero linear coefficient. Both $F - f$ and $F \circ g - f \circ g$ are analytic. Lemma 2.2 therefore transfers the equality from F to f . Finally apply Proposition 5.1 to the analytic outer function translated to the constant term $f(0)$. $\square$

The theorem does not assert a type formula for composition of two arbitrary divergent series. Analyticity of the coordinate maps is an essential part of the statement proved here. It also does not assert invariance under changes with zero derivative, such as ramification $z \mapsto z^d$; those changes rescale coefficient indices rather than merely the exponential type.

6 Gevrey and logarithmically refined weights

6.1 A family extending all positive Gevrey orders

For real parameters s, β , define for $n \geq 1$

$$W_n(s, \beta) = (n!)^s (\log(n + e))^{\beta n}, \quad W_0 = 1. \quad (6.1)$$

The logarithmic shift ensures positivity at small indices; changing it does not affect any type considered below.

Lemma 6.1 (Admissibility of the refined weights). *Suppose either $s > 0$ and $\beta \in \mathbb{R}$, or $s = 0$ and $\beta > 0$. After changing only finitely many initial terms, (6.1) is a log-convex superexponential weight. Moreover,*

$$W_n(s, \beta)^{1/n} \sim (n/e)^s (\log n)^\beta. \quad (6.2)$$

Proof. Write $F_n = \log W_n$ and $B(x) = x \log \log(x + e)$. Then

$$F_{n+1} - 2F_n + F_{n-1} = s \log \frac{n+1}{n} + \beta(B(n+1) - 2B(n) + B(n-1)).$$

Elementary differentiation gives

$$B''(x) = \frac{1}{x \log x} \left(1 + O((\log x)^{-1}) \right).$$

The discrete second difference of B is the integral of B'' against a nonnegative triangular kernel of total mass one on $[n-1, n+1]$. Consequently the second difference of F_n is

$$\frac{s}{n} + O(n^{-2}) + \frac{\beta}{n \log n} \left(1 + O((\log n)^{-1}) \right),$$

which is eventually positive under either hypothesis. Moreover, using Stirling's formula only for $\log(n!)$, for $s > 0$ one gets

$$F_{n+1} - F_n - \frac{F_n}{n} = s + o(1) > 0$$

for sufficiently large n . For $s = 0$, $\beta > 0$, the eventual convexity of B and the exact identity

$$B'(x) - \frac{B(x)}{x} = \frac{x}{(x + e) \log(x + e)} > 0$$

give the same positivity directly. Thus there is an integer N in the convex tail such that

$$\frac{F_N}{N} \leq F_{N+1} - F_N.$$

Keep W_n for $n \geq N$, and set $\widetilde{M}_n = W_N^{n/N}$ for $0 \leq n < N$. The initial logarithms lie on a straight line; the displayed inequality makes their join with the convex tail convex. Thus $\widetilde{M}$ is a globally log-convex weight with the same tail. Stirling's formula proves (6.2), which tends to infinity under the stated hypotheses. $\square$

Corollary 6.2 (Exact refined-type inversion). *Under either hypothesis in Lemma 6.1, let*

$$T_{s,\beta}(f) = \limsup_n \left(\frac{|[z^n]f|}{(n!)^s \log(n + e)^{\beta n}} \right)^{1/n}.$$

For every formal diffeomorphism with derivative $a \neq 0$,

$$T_{s,\beta}(f^{-1}) = \frac{T_{s,\beta}(f)}{|a|}. \quad (6.3)$$

The analytic coordinate law (5.3) holds for these refined types as well.

Proof. Use the finite modification from Lemma 6.1 in Theorems 4.1 and 5.2. Finite changes of weights do not alter type. $\square$

For $s > 0$, $\beta = 0$, this gives exact Gevrey-type preservation, not merely closure of the Gevrey class. For $s = 0$, $\beta > 0$, it gives the same exact preservation on a scale slower than every positive Gevrey weight. By contrast, $s = \beta = 0$ is the analytic endpoint excluded by Theorem 4.2.

6.2 Size of the explicit overhead

Fix an admissible globally log-convex modification and a bound (3.4). From (3.7) and (6.2),

$$\log \frac{|b_n|}{A^{n-1} M_n} \leq O(1) + O\left(n^{1-s/2} (\log n)^{-\beta/2}\right), \quad (6.4)$$

with the convention that a zero coefficient satisfies the inequality. The constants depend on the fixed input bound and the finite weight modification. In the sub-Gevrey case $s = 0$, $\beta > 0$, the overhead is $O(n/(\log n)^{\beta/2})$, still $o(n)$.

This is a *proved upper bound*, not a sharp asymptotic for nonlinear inverse corrections. Its role is to retain the exponential constant, and its optimization is a separate research question.

7 Application to countable exponential feedback

7.1 Formal existence and a positive coefficient formula

Fix finite real numbers $\lambda_j \geq 0$, without any growth assumption. Consider

$$\Phi(q, u) = \sum_{j \geq 1} q^j e^{\lambda_j u}, \quad \widehat{U}(q) = \Phi(q, \widehat{U}(q)). \quad (7.1)$$

The coefficient of q^n on the right depends only on coefficients of $\widehat{U}$ below degree n . Therefore the equation determines a unique series $\widehat{U} \in q\mathbb{R}[[q]]$, with $u_1 = 1$ if $\widehat{U} = \sum_{n \geq 1} u_n q^n$. The countable kernel need not converge in a complex neighborhood of $(0, 0)$; formal existence does not require such convergence.

Proposition 7.1 (Finite positive Lagrange formula). *Let $\mathbf{m} = (m_1, m_2, \dots)$ have finite support, and write $|\mathbf{m}| = \sum m_j$, $w(\mathbf{m}) = \sum j m_j$. Then*

$$u_n = \sum_{w(\mathbf{m})=n} \frac{(\sum_j \lambda_j m_j)^{|\mathbf{m}|-1}}{\prod_j m_j!}, \quad n \geq 1, \quad (7.2)$$

where $0^0 = 1$. In particular $u_n \geq 1$.

Proof. Introduce a marker t and solve $V = t\Phi(q, V)$ formally. Lagrange inversion in t gives

$$[t^k]V = \frac{1}{k} [u^{k-1}] \Phi(q, u)^k.$$

Every term has q -order at least k , so setting $t = 1$ is coefficientwise well-defined. For an ordered k -tuple $(j_1, \dots, j_k)$, the contribution is

$$\frac{1}{k!} (\lambda_{j_1} + \dots + \lambda_{j_k})^{k-1} q^{j_1 + \dots + j_k}.$$

Grouping tuples by their multiplicities gives (7.2). All terms are nonnegative, and the tuple of length one with entry n contributes one. $\square$

With $q = e^{-x}$, the formal series becomes $\sum_{n \geq 1} u_n e^{-nx}$. It already has a simple well-based one-generator exponential support. The questions below concern the coefficient growth on that support, not a failure of support well-ordering. Since $u_1 = 1$, the inverse $\widehat{Q}(u) = \sum_{n \geq 1} q_n u^n$ exists formally and satisfies the literal formal identity

$$u = \sum_{j \geq 1} \widehat{Q}(u)^j e^{\lambda_j u}. \quad (7.3)$$

7.2 The two forward inputs

The following are the precise results taken from [3]. They are labeled as inputs to avoid attributing their proofs to the present article.

Repository input 7.2 (Exact forward Gevrey type). *For the nonnegative-slope model (7.1) and every $s > 0$,*

$$T_s(\widehat{U}) := \limsup_n \left(\frac{u_n}{(n!)^s} \right)^{1/n} = \left(\frac{s+1}{s} \right)^{s+1} \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^{s+1}}. \quad (7.4)$$

The equality holds in $[0, \infty]$.

Repository input 7.3 (Regularly varying forward slopes). *Suppose $a > 0$, $p > 1$, $\beta \in \mathbb{R}$, and $\lambda_j \sim a j^p (\log j)^\beta$. Then*

$$\begin{aligned} \log u_n &= (p-1)n \log n + (\beta-p)n \log \log n \\ &\quad + \left(\log a + p \log \frac{p}{p-1} - (p-1) \right) n + o(n). \end{aligned} \quad (7.5)$$

In the cited source these are the statements labeled `thm:classification`, equation `eq:exacttype`, and `thm:regular`, equation `eq:logasymptotic`, respectively. They are conventional research results, not formalized statements. Neither our abstract inverse theorem nor its coordinate-change theorem uses these inputs; the remaining feedback applications do.

7.3 The sharp inverse classification

Theorem 7.4 (Exact inverse Gevrey type). *For (7.1) and its formal inverse, for every $s > 0$,*

$$T_s(\widehat{Q}) = T_s(\widehat{U}) = \left(\frac{s+1}{s}\right)^{s+1} \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^{s+1}}. \quad (7.6)$$

Consequently $\widehat{Q}$ belongs to Gevrey class s exactly when $\lambda_j = O((j \log(j+1))^{s+1})$, and its exact type is the constant in (7.6).

Proof. For $M_n = (n!)^s$, the weight is log-convex and superexponential. Theorem 4.1 gives $T_s(\widehat{Q}) = T_s(\widehat{U})$ because $u_1 = 1$. Insert Repository input 7.2. A finite type is equivalent to a bound $CA^n(n!)^s$, and the limsup on the right is finite exactly under the stated big- O condition, since every individual slope is finite. $\square$

The class-membership equivalence was already available in [3]; the addition in (7.6) is the exact constant for the inverse, including the zero and infinite cases. Nonnegativity is an assumption on the primitive slopes and the imported forward results, not on the inverse coefficients.

Corollary 7.5 (Exact inverse Borel radius). *Define the integrated ordinary Borel series by*

$$(\mathcal{B}\widehat{Q})(\zeta) = \sum_{n \geq 1} \frac{q_n}{n!} \zeta^n. \quad (7.7)$$

Writing

$$A_* := \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^2},$$

its exact radius of convergence is

$$R_{\mathcal{B}\widehat{Q}} = \frac{1}{4A_*}, \quad (7.8)$$

with $1/0 = \infty$ and $1/\infty = 0$. More generally, the series $\sum q_n \zeta^n / (n!)^s$ has radius $1/T_s(\widehat{Q})$.

Proof. Apply Cauchy–Hadamard to (7.7), followed by Theorem 7.4 with $s = 1$. The general assertion is the same root test with the stated normalization. $\square$

For example, if $\lambda_j \sim A(j \log j)^2$ with $A > 0$, the exact radius is $1/(4A)$. If $\lambda_j = j^2$, the Borel transform is entire. If $A_* = \infty$, it has zero radius as a formal Borel series. A finite positive radius implies at least one obstruction to analyticity on the boundary circle, but does not identify its angle or its local nature. An infinite radius says nothing by itself about a Laplace integral on any prescribed ray. For nonintegral s , the factorial normalization above is the one explicitly defined here, and should not be confused with a different k -summability convention involving $\Gamma(1 + sn)$.

7.4 A sharp logarithmically refined inverse invariant

Theorem 7.6 (Regularly varying inverse type). *Under Repository input 7.3, put*

$$s = p - 1, \quad \gamma = \beta - p, \quad K = a \left(\frac{p}{p-1} \right)^p. \quad (7.9)$$

Then

$$\limsup_{n \rightarrow \infty} \left(\frac{|q_n|}{(n!)^{p-1} \log(n+e)^{(\beta-p)n}} \right)^{1/n} = K. \quad (7.10)$$

For the forward coefficients, the corresponding normalized n th roots have an actual limit equal to K .

Proof. Stirling's formula gives

$$\log((n!)^{p-1} \log(n+e)^{(\beta-p)n}) = (p-1)n \log n - (p-1)n + (\beta-p)n \log \log n + o(n).$$

Subtract this from (7.5) and divide by n . The resulting limit is $\log K$. Thus the forward refined type is K , and in fact the forward normalized roots converge to it. The weight satisfies Lemma 6.1 because $p-1 > 0$. Corollary 6.2 transfers its exact type to $\hat{Q}$. $\square$

An immediate consequence is a two-sided exponential envelope in the limsup sense: for each $\varepsilon > 0$,

$$|q_n| \leq (K + \varepsilon)^n (n!)^{p-1} \log(n+e)^{(\beta-p)n}$$

for all sufficiently large n ; if $0 < \varepsilon < K$, the reversed bound with $K - \varepsilon$ holds at infinitely many n . Modifying finitely many slopes does not change this invariant when the same regularly varying tail is retained. The theorem does not say that such a modification preserves each coefficient or each analytic summation.

Corollary 7.7 (Quadratic feedback). *For $\lambda_j = j^2$,*

$$\limsup_{n \rightarrow \infty} \left(\frac{|q_n| \log(n+e)^{2n}}{n!} \right)^{1/n} = 4. \quad (7.11)$$

Both $\hat{U}$ and $\hat{Q}$ have Gevrey-one type zero, belong to no Gevrey class $s < 1$ with $s > 0$, and have entire ordinary Borel transforms.

Proof. Use $a = 1, p = 2, \beta = 0$ in Theorem 7.6. The exact ordinary Gevrey types also follow from (7.6): the slope ratio tends to zero for $s = 1$ and to infinity for $0 < s < 1$. The Borel conclusion follows from Corollary 7.5. $\square$

8 Entire Borel growth from the refined type

8.1 A coefficient-to-growth lemma that permits cancellation

For an entire function F , write $\mathcal{M}_F(r) = \max_{|z|=r} |F(z)|$. In iterated logarithms, replacing $\mathcal{M}_F(r)$ by $\max(e, \mathcal{M}_F(r))$ at small radii makes all expressions defined and does not affect any limit below.

Lemma 8.1 (Iterated-logarithmic growth). *Suppose $F(z) = \sum_{n \geq 0} c_n z^n$, $d > 0$, $K > 0$, and*

$$\limsup_{n \rightarrow \infty} |c_n|^{1/n} \log(n+e)^d = K. \quad (8.1)$$

Then F is entire and

$$\limsup_{r \rightarrow \infty} \frac{\log \log \mathcal{M}_F(r)}{r^{1/d}} = K^{1/d}. \quad (8.2)$$

In particular F has infinite classical entire order.

Proof. Equation (8.1) implies $\limsup |c_n|^{1/n} = 0$, so F is entire. For an upper bound choose $L > K$ and $\eta > 0$. For all sufficiently large n ,

$$|c_n| \leq \frac{L^n}{\log(n+e)^{dn}}.$$

For sufficiently large r , set

$$N = \left\lceil \exp((1+\eta)(Lr)^{1/d}) \right\rceil.$$

For $n \geq N$, one has $Lr/\log(n+e)^d \leq (1+\eta)^{-d} =: \rho < 1$. The tail is at most $\sum_{n \geq N} \rho^n$ in absolute value on $|z| = r$. The terms before N are at most a fixed polynomial in r plus $N \max(1, Lr)^N$. Consequently

$$\log \log \mathcal{M}_F(r) \leq (1+\eta)(Lr)^{1/d} + O(\log \log r).$$

Taking the limsup and then $L \downarrow K$, $\eta \downarrow 0$ gives the upper bound in (8.2).

For the lower bound, choose $0 < \varepsilon < K$ and $\eta > 0$. There are infinitely many n for which

$$|c_n| \geq \frac{(K - \varepsilon)^n}{\log(n+e)^{dn}}.$$

At the radii

$$r_n = \frac{e^{d\eta} \log(n+e)^d}{K - \varepsilon},$$

Cauchy's coefficient inequality gives $\mathcal{M}_F(r_n) \geq |c_n| r_n^n \geq e^{d\eta n}$. It follows that

$$\limsup_{r \rightarrow \infty} \frac{\log \log \mathcal{M}_F(r)}{r^{1/d}} \geq e^{-\eta} (K - \varepsilon)^{1/d}.$$

Let $\eta \downarrow 0$ and $\varepsilon \downarrow 0$. The positive finite value in (8.2) implies $\limsup \log \log \mathcal{M}_F(r) / \log r = \infty$, which is infinite classical order. $\square$

No positivity assumption is present in this lemma. The lower bound uses Cauchy's inequality for a coefficient, rather than a value of the function on a prescribed ray. This distinction is particularly important for a signed inverse series.

8.2 Feedback transforms of integer factorial order

Theorem 8.2 (Exact inverse Borel maximum-modulus type). *Assume*

$$\lambda_j \sim a j^p (\log j)^\beta, \quad p = m + 1, \quad m \in \mathbb{N}, \quad m \geq 1, \quad a > 0, \quad \beta < p.$$

Set $d = p - \beta > 0$ and $K = a(p/(p-1))^p$. Define

$$B_{m,Q}(z) = \sum_{n \geq 1} \frac{q_n}{(n!)^m} z^n, \quad B_{m,U}(z) = \sum_{n \geq 1} \frac{u_n}{(n!)^m} z^n. \quad (8.3)$$

Both are entire, and each satisfies

$$\limsup_{r \rightarrow \infty} \frac{\log \log \mathcal{M}_B(r)}{r^{1/d}} = K^{1/d}. \quad (8.4)$$

Proof. Theorem 7.6, with $m = p - 1$, says that the coefficients of $B_{m,Q}$ satisfy (8.1). For $B_{m,U}$ the normalized root even has a limit, so the same hypothesis holds. Apply Lemma 8.1 to each transform. $\square$

For $m = 1$ this is the ordinary integrated Borel transform; for larger integer m it is the explicitly defined repeated factorial normalization, not an unstated alternative convention. In the quadratic case, (8.4) becomes

$$\limsup_{r \rightarrow \infty} \frac{\log \log \mathcal{M}_{\widehat{BQ}}(r)}{\sqrt{r}} = 2. \quad (8.5)$$

The coefficient information proved here justifies a *limsup* for the inverse. It does not justify replacing that limsup by a limit or $\mathcal{M}_{\widehat{BQ}}(r)$ by $|(\widehat{BQ})(r)|$. In particular, infinite maximum-modulus order does not contradict summability on a direction where the same entire function grows much more slowly. This is consistent with the separate negative-direction result in [4].

Editorial note (ProveIt, 2026-09-29). For $\beta = 0$ the forward half of Theorem 8.2 is Theorem `thm:borelintro` of [3] (`Exponential_Feedback_Regularity_Classification/article.tex`, lines 301–316), with the same constant $K^{1/p} = a^{1/p}p/(p-1)$; for these positive coefficients that theorem even gives a limit along the positive ray, which is the maximum modulus. The new content is the inverse transform $B_{m,\widehat{Q}}$ and the range $\beta \neq 0$. For $\lambda_j = aj^2$ two later packages (batch 49) claim a limit for the inverse: [11], Theorem `thm:borel`, gives $-(\mathcal{B}\widehat{Q})(R) \sim \mathcal{M}_{\mathcal{B}\widehat{Q}}(R)$ with an explicit equivalent and $\log \log \mathcal{M}_{\mathcal{B}\widehat{Q}}(r)/\sqrt{r} \rightarrow 2\sqrt{a}$ (the value 2 of (8.5) when $a = 1$), and [10], Theorem `thm:borel`, gives the same positive-ray equivalent. Both are unreviewed and consistent with the limsup proved here.

9 Non-D-finiteness and analytic-normal-form obstructions

9.1 Two elementary holonomic facts

Definition 9.1. A formal series $F(z)$ is D-finite if it satisfies a nonzero linear differential equation

$$p_r(z)F^{(r)}(z) + \cdots + p_0(z)F(z) = 0, \quad p_j \in \mathbb{C}[z], \quad p_r \neq 0. \quad (9.1)$$

A sequence is P-recursive if, for all sufficiently large n , it satisfies a nonzero linear recurrence with polynomial coefficients in n .

Lemma 9.2 (Factorial division preserves D-finiteness). *If $F(z) = \sum a_n z^n$ is a D-finite formal series and m is a nonnegative integer, then $\sum a_n z^n / (n!)^m$ is D-finite formally.*

Proof. Coefficient extraction in (9.1) gives a polynomial-coefficient recurrence for a_n , after shifting its indices to be nonnegative. Write it as

$$\sum_{j=0}^r P_j(n) a_{n+j} = 0$$

for all sufficiently large n , with not all P_j zero. Put $a_n = (n!)^m c_n$ and divide by $(n!)^m$. This gives

$$\sum_{j=0}^r P_j(n) ((n+1) \cdots (n+j))^m c_{n+j} = 0.$$

Every coefficient is polynomial because m is an integer, and the recurrence is nonzero. Such a recurrence yields a linear differential equation for $\sum c_n z^n$ by multiplying by powers of z , summing in n , and replacing n by the Euler operator $z d/dz$. Finitely many exceptional initial indices contribute a polynomial inhomogeneous term, which is annihilated by a sufficiently high derivative. This yields a nonzero homogeneous linear differential operator, proving D-finiteness. $\square$

Lemma 9.3 (Entire D-finite functions have finite order). *Every entire D-finite function has finite classical entire order.*

Proof. The zero function is harmless. A nonzero entire solution has an equation (9.1) of order $r \geq 1$. Choose R so large that all zeros of p_r lie in $|z| < R$. Outside that disk, the vector

$$Y = (F, F', \dots, F^{(r-1)})^\top$$

satisfies a companion system $Y' = A(z)Y$ whose entries are rational functions. There exist $C > 0$ and an integer $L \geq 0$ with $\|A(z)\| \leq C|z|^L$ for $|z| \geq R$. The initial values $Y(Re^{i\theta})$ are uniformly bounded in θ , since all components are entire. Along a radial segment, Gronwall's inequality therefore gives, uniformly in θ ,

$$\|Y(re^{i\theta})\| \leq C_0 \exp \left(C \int_R^r t^L dt \right) \leq C_0 \exp(C_1 r^{L+1}).$$

Thus $\mathcal{M}_F(r) \leq C_0 \exp(C_1 r^{L+1})$, which is finite classical order. $\square$

9.2 An obstruction valid for signed coefficient sequences

Theorem 9.4 (Logarithmic factorial obstruction). *Let $m \geq 1$ be an integer, $d > 0$, and $K > 0$. Suppose a formal series $F(z) = \sum a_n z^n$ satisfies*

$$\limsup_{n \rightarrow \infty} \left(\frac{|a_n| \log(n+e)^{dn}}{(n!)^m} \right)^{1/n} = K. \quad (9.2)$$

Then F is not D-finite, and its coefficient sequence is not P-recursive.

Proof. Lemma 8.1 says that $B(z) = \sum a_n z^n / (n!)^m$ is entire and of infinite order. If F were D-finite, Lemma 9.2 would make B D-finite as a formal series. Since B is entire, its formal differential identity holds analytically by termwise differentiation near zero and then by the identity theorem. Lemma 9.3 gives a contradiction. The equivalence between D-finite series and P-recursive coefficients follows from the coefficient/operator translations in the proof of Lemma 9.2; hence the second conclusion. $\square$

Corollary 9.5 (Non-D-finiteness of feedback and its inverse). *Under the hypotheses of Theorem 8.2, neither $\hat{U}$ nor $\hat{Q}$ is D-finite. In particular, for $\lambda_j = j^2$ neither series satisfies a nonzero linear differential equation with polynomial coefficients, and neither coefficient sequence is P-recursive.*

Proof. Apply Theorem 9.4 to the forward and inverse refined types given by Theorem 7.6, with $d = p - \beta$. $\square$

Corollary 9.6 (No analytic change to a D-finite normal form). *Let F satisfy (9.2). Let g be analytic with $g(0) = 0$, $g'(0) \neq 0$, and let H be analytic near $F(0)$ with $H'(F(0)) \neq 0$. Then $H \circ F \circ g$ is not D-finite.*

Proof. Apply Theorem 5.2 to the weight $(n!)^m \log(n+e)^{-dn}$, with a finite log-convex modification as in Lemma 6.1. The transformed type is $|g'(0)|K$, which is positive and finite. Thus the transformed series again satisfies the hypothesis of Theorem 9.4. $\square$

The last corollary is a growth obstruction, not a claimed general closure property of the D-finite class under arbitrary analytic composition. It says that the particular divergent growth profile cannot be removed by invertible analytic coordinate changes.

9.3 Limits of the obstruction

Non-D-finiteness does *not* mean differential transcendence. A series can fail every linear polynomial-coefficient differential equation and still satisfy a nonlinear algebraic differential equation. No exclusion of such nonlinear equations is proved here.

The logarithmic deficit $d > 0$ is also essential to this proof. At $\beta = p$, the normalized Borel series has a finite positive radius rather than being entire. For $\beta > p$, the same normalization has zero radius. Lemma 9.3 cannot be applied in either case. Likewise, for noninteger $p - 1$, the polynomial-recurrence argument for dividing by $(n!)^{p-1}$ is unavailable as stated. These are boundaries of the present theorem, not assertions that the series at those boundaries are D-finite.

10 Exact computations and reproducibility

10.1 Independent finite constructions

The accompanying program `code/verify.py` uses only the Python standard library. All identity and inequality checks use `fractions.Fraction`; decimal diagnostics are generated separately and labeled as floating point.

The forward coefficients are computed by a triangular exponential recurrence. If

$$E_j(q) = \exp(\lambda_j \widehat{U}(q)) = \sum_{r \geq 0} E_{j,r} q^r,$$

then

$$E_{j,0} = 1, \quad E_{j,r} = \frac{\lambda_j}{r} \sum_{k=1}^r k u_k E_{j,r-k}, \quad u_n = \sum_{j=1}^n E_{j,n-j}. \quad (10.1)$$

This construction is separate from the inverse construction, which uses (3.2). Both inverse compositions are subsequently checked by direct polynomial multiplication. Equation (7.3) is checked independently by substituting the computed inverse into the literal finite truncation of the feedback kernel.

The finite majorant (3.5) is tested on signed rational input series for four different log-convex weights: $n!$, $(n!)^2$, 2^{n^2} , and a slowly growing weight with nondecreasing quotients $1 + \text{bitlength}(n)$. That last choice tests the mechanism beyond factorial growth without using transcendental arithmetic. The geometric-mean weight inequalities are checked after raising to integer powers, avoiding numerical roots.

10.2 Recorded results

The recorded command was

```
python code/verify.py -order 70
```

and it completed successfully. The exact report contains:

- Forward and inverse quadratic coefficients through degree 70; both inverse compositions, the literal inverse kernel, linear rescaling, and the Catalan endpoint example verified through degree 30.
- Twelve seeded signed-series cases, with 180 exact checks of the finite inverse majorant and 8 188 exact concentration inequalities.
- Geometric-mean weight checks for four weights through degree 12, and an analytic postcomposition/inverse factorization through degree 30.

The random seed is 20260929. The program raises an exception on a failed check; it does not silently convert failed tests into warnings. Table 2 gives the first ten exact coefficients.

These tests cannot verify a limsup over all indices or a statement about entire growth at infinite radius. In particular, the floating-point refined-root diagnostics at degree 70 are approximately 7.59947 for $\widehat{U}$ and 6.75803 for $\widehat{Q}$, not close to the asymptotic invariant 4. This slow preasymptotic behavior is recorded rather than hidden. No monotone convergence claim or extrapolation of the constant is based on that table. The asymptotic conclusions depend on the proofs and on the two stated forward inputs, not on finite numerical proximity.

n	$u_n = [q^n]\widehat{U}(q)$	$q_n = [u^n]\widehat{Q}(u)$
1	1	1
2	2	-2
3	15/2	1/2
4	107/3	-2/3
5	1555/8	-29/8
6	17362/15	-743/60
7	5288311/720	-8491/144
8	15414767/315	-92726/315
9	2734575197/8064	-611063/384
10	983674676/405	-1667615261/181440

Table 2: Quadratic feedback $\lambda_j = j^2$: exact rational coefficients.

10.3 Contents and build procedure

The package contains the article source and PDF, the verification program, exact coefficient data, numerical diagnostics, the recorded verification report, a source-provenance file, and a Makefile. The TeX source includes an embedded bibliography and the exact coefficient table generated by the program. It compiles independently of the data directory and has no network dependency.

To rebuild from the package directory, run

```
python code/verify.py -order 70
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verification program does not modify ProveIt or contact external services. The emitted `verification.json` distinguishes exact checks from non-certified floating-point diagnostics. PDF layout checks and build status are recorded separately in `data/build_validation.json`.

11 Further research questions and a formalization program

The following questions are proposed by this article. They are not all claimed to be previously published open problems. Each specifies a concrete next theorem or obstruction to investigate.

11.1 Optimal subexponential cost of reversion

The bound $\exp\{O(n/\sqrt{r_n})\}$ is sufficient for exact type but comes from deliberately coarse composition counts. What is the smallest universal overhead for a fixed weight and fixed input constants? Can one replace the square-root loss by an expression involving the local quotients M_n/M_{n-1} , and construct signed or nonnegative examples that attain it? A useful first target is $M_n = (n!)^s$ with $0 < s < 1$, where many small nonlinear blocks may still contribute a visible subexponential factor.

Editorial note (ProveIt, 2026-09-29). For $M_n = (n!)^s$, and for the shifted weight $((n-1)!)^s$, the later package [13] (batch 52; not independently reviewed) finds the worst case over the whole coefficient ball $|a_n| \leq CA^{n-1}((n-1)!)^s$: one negative saturating series is extremal in every degree, its normalized inverse amplification R_n satisfies $\log R_n \sim Cn^{1-s}$ for $0 < s < 1$ and $R_n \sim \exp(Cn^{1-s})$ for $s > 1/2$, with an extra constant at $s = 1/2$, and the fixed ball is stable

under inversion exactly for $s \geq 1$. So the overhead $\exp\{O(n/\sqrt{r_n})\}$ of Theorem 3.1 is not sharp. The local-quotient version for general weights remains open.

11.2 Full inverse asymptotics for quadratic feedback

For $\lambda_j = j^2$, determine whether the normalized inverse roots in (7.11) actually converge to four. A stronger target is an asymptotic equivalent for q_n , including its sign and subexponential amplitude. The small-order sign pattern in Table 2 is not a proof of an eventual sign law. The two-sided inverse argument used here deliberately avoids that issue; a complete answer likely requires control of cancellation between Lagrange blocks.

Editorial note (ProveIt, 2026-09-29). The stronger target is Conjecture `conj:quadratic-inverse` of [9] (`Finite_Core_Universality_Exponential_Feedback/article.tex`, lines 1704–1710): $q_n \sim -S_n \exp(-(1 + 1/a)r_n)$ for $\lambda_j = aj^2$, with $r_n(1 + r_n)e^{2r_n} = an$ and the explicit saddle quantity S_n defined there. This article neither proves nor contradicts it, since the factors separating it from the forward equivalent are $e^{o(n)}$. Two later packages (batch 49) each claim a proof, independently and under different hypotheses: [10], Theorem `thm:main`, and [11], Theorem `thm:main`; neither has been independently reviewed. If correct, they answer both parts of this question for $\lambda_j = j^2$: the inverse coefficients are eventually negative with that equivalent, and [11], Corollary `cor:gevrey`, gives $\log(|q_n|/n!) = -2n \log \log n + n \log(4a) + o(n)$, so the normalized roots in (7.11) converge to 4.

11.3 Angular Borel growth and summation

Determine the directional growth of $\widehat{\mathcal{BQ}}(re^{i\theta})$ rather than its maximum over the circle. Theorem 8.2 supplies an exact global limsup scale, while [4] supplies a different, directional summation result. Can these be joined into an angular indicator theory, including the transition between the left and right half-planes? This is also a route toward testing angular rather than fine tubular summability.

11.4 Critical Borel singularities

For $\lambda_j \sim A(j \log j)^2$, the radius $1/(4A)$ is now fixed for the inverse. Which boundary points are singular, what are their local singularity types, and how do they depend on lower-order slope terms? Positivity of the forward coefficients helps locate a forward singularity, but cannot be transferred to a signed inverse without a separate argument. A radius theorem is not yet a Stokes-constant theorem.

11.5 Weights beyond log-convexity

Theorem 4.2 gives a complete answer *within* the log-convex class. Find necessary and sufficient conditions for arbitrary positive weights to have the same exact reversion invariant. A possible approach is to compare a weight with a log-convex envelope up to $\exp(o(n))$ factors. The difficult cases are oscillatory weights whose deep dips occur precisely at indices populated by nonlinear compositions.

11.6 Two divergent composition arguments

Analytic pre- and postcomposition are controlled exactly here. Determine what can be said when both series are divergent. A naive equality in terms of the two types alone cannot ignore cancellations, since a series composed with its inverse is analytic. A useful replacement may be

a filtered composition estimate together with a noncancellation condition, or a two-parameter profile that retains the position of large coefficients.

11.7 Multivariate and operator-valued inversion

Extend the zero-loss theorem to formal maps in several variables, where the derivative is an invertible matrix and homogeneous degrees have many monomials. The natural question is whether an exact type transforms by a directional norm of the inverse Jacobian rather than a single scalar. A Banach-space formulation should distinguish constants coming from multilinear operator norms from combinatorial losses that are only subexponential in the degree.

Editorial note (ProveIt, 2026-09-29). The later package [12] (batch 52; conventional proofs, not independently reviewed) answers “Weights beyond log-convexity”: for an arbitrary positive weight, with the shift $N_n = M_{n+1}$, tangent reversion preserves the type universally exactly when $N_n^{1/n} \rightarrow \infty$ and $\log(S_n/N_n) = o(n)$, where S_n is the largest product $N_{j_1} \cdots N_{j_k}$ over compositions of n ; when $N_n^{1/n} \rightarrow \infty$, $\limsup(S_n/N_n)^{1/n}$ is the optimal distortion. A log-convex envelope up to $e^{o(n)}$ is not necessary. It also shows that the equivalence (ii) $\Leftrightarrow$ (iii) of Theorem 4.2 needs the log-convexity assumed there. For “Two divergent composition arguments” it proves $\tau_M(f \circ g) \leq \max\{\tau_M(f), \tau_M(g)\}$, with equality for unequal types and every value in $[0, T]$ possible at equal types T ; the factorial case was proved independently, by a different majorant, in [13]. For “Multivariate and operator-valued inversion” it proves the tangent case on Banach spaces without dimension loss and sharp two-sided bounds $\tau_M(F)/\|A\| \leq \tau_M(F^{-1}) \leq \|A^{-1}\| \tau_M(F)$ for a general derivative A , which a single scalar type cannot refine; a directional theory remains open.

11.8 Hahn supports and accumulating action sets

For a well-based exponential support not identified with the positive integers, choose a meaningful replacement for degree and for M_n . Can an action filtration support an analogue of the concentration inequality? If infinitely many primitive actions lie below a finite cut, the finite ordered-composition count used in Theorem 3.1 is no longer available verbatim. A theorem there must explicitly separate support summability from coefficient majorization.

11.9 Differential transcendence and the critical boundary

Strengthen Corollary 9.5 from a linear obstruction to an obstruction against nonlinear algebraic differential equations, or find a nonlinear equation that the feedback model does satisfy. In parallel, determine non-D-finiteness at $\beta \geq p$, where the entire normalization argument fails, and for noninteger $p - 1$. These are separate questions; none is settled merely by the terminology “non-holonomic.”

11.10 Summation compatible with refined inversion

For logarithmic Gevrey deficits, identify a moment or acceleration procedure that matches the full weight rather than just its factorial order. Under what analytic continuation and growth hypotheses does its sum commute with inversion and retain the exact type? Such a theorem would have to add analytic hypotheses absent from the purely formal result proved here, and must not infer summability from a zero type.

11.11 A staged Lean formalization target

A practical formalization can be divided into four theorem-sized stages. First prove the finite log-convex concentration inequalities and the geometric-mean ratio bound over positive real sequences. Next formalize (3.3) and (3.5) for coefficients of an inverse formal power series, with the sum over positive compositions made explicit. Third obtain the subexponential bound and extended-real limsup theorem, including zero and infinite cases and the Catalan necessity example. Finally connect the result to the particular feedback construction and formally prove or import verified versions of Repository inputs 7.2 and 7.3.

The general theorem is independent of those last two inputs, so it is a useful standalone milestone. The finite exact checks in this package supply examples and regression tests, not substitutes for those Lean proofs.

12 Conclusion

The central result is a complete zero-loss criterion for log-convex coefficient weights: universal exact type preservation under formal reversion is equivalent to superexponential growth of the weight roots. The proof provides a finite majorant rather than only an existence statement, and yields exact analytic-coordinate covariance without moderate-growth or summability assumptions.

Applied to the stated forward feedback estimates, this resolves the repository's sharp inverse-type gap. It fixes the inverse Borel radius, preserves logarithmically refined constants despite signed cancellation, and yields a non-D-finiteness obstruction for an integer-order family, including quadratic feedback. The next genuinely unresolved steps are finer inverse asymptotics, angular Borel growth, and extensions beyond log-convex degree weights; they are not concealed inside the proved limsup statements.

A A formal residue derivation of Lagrange reversion

For completeness, let $f(t) = at + O(t^2)$, $a \neq 0$, and let $g = f^{-1}$. For a formal Laurent series with a finite principal part, denote by res_t the coefficient of t^{-1} . Formal substitution gives

$$\text{res}_z A(z) = \text{res}_t A(f(t))f'(t). \quad (\text{A.1})$$

It is enough to check monomials $A(z) = z^k$. For $k \neq -1$ the right side is the residue of $(f^{k+1})'/(k+1)$ and is zero. For $k = -1$, write $f = at(1+h)$; then $f'/f = t^{-1} + h'/(1+h)$ has residue one. At each coefficient only finitely many relevant terms enter, justifying linear extension.

For $n \geq 1$, apply (A.1) to $A(z) = g(z)z^{-n-1}$:

$$[z^n]g(z) = \text{res}_t t f'(t) f(t)^{-n-1}.$$

The residue of the derivative of $t f(t)^{-n}$ is zero. Since

$$(t f^{-n})' = f^{-n} - n t f' f^{-n-1},$$

we obtain

$$[z^n]g(z) = \frac{1}{n} \text{res}_t f(t)^{-n} = \frac{1}{n} [t^{n-1}] \left(\frac{t}{f(t)} \right)^n. \quad (\text{A.2})$$

This gives (3.2) at $a = 1$ and also makes the scaling factor a^{-n} visible. The usual marker form $V = t\Phi(V)$ follows by putting $f(v) = v/\Phi(v)$ when $\Phi(0)$ is a unit. For Proposition 7.1, work first over $\mathbb{C}((q))$: the constant $\Phi(q, 0) = q/(1-q)$ is a unit in this Laurent-series field, so the argument

applies verbatim to inversion in the marker variable. Its displayed coefficient formula has q -order at least k in marker degree k , and therefore descends to the coefficientwise specialization at marker value one in $\mathbb{C}[[q]]$. Thus the calculation remains formal even when the resulting series diverges everywhere except zero.

B Pinned source record and scope of novelty review

The snapshot is 0c973d8f5e7ef4550f4df1bf486d890037fd9f14, timestamped 29 September 2026, 20:14:16 UTC. Repository bibliography links are pinned to that commit. The project README and these files under `Analysis/Transseries/docs/series-and-transseries/` were the principal sources:

```
Exponential_Feedback_Regularity_Classification/article.tex
Exponential_Feedback_Regularity_Classification/README.md
Negative_Ray_Summation_Exponential_Feedback/README.md.
```

The first predecessor retains older internal snapshot references from before the directory migration; those are not substituted for the snapshot above. No upstream files were changed. Full links, retrieval scope, and dependency boundaries are recorded in `PROVENANCE.json`.

The documented boundary resolved here is exact inverse type, not merely Gevrey membership. This does not establish absence of every result from all previous publications or concurrent notes. The classical ingredients are Lagrange inversion, log-convex majorization, root tests, and D-finite coefficient translations. The proposed contribution is their zero-loss combination, the sharp weight threshold, and the feedback consequences.

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- [2] ProveIt contributors, *Transseries*, project README, same snapshot. [Pinned project README](#).
- [3] *A sharp regularity classification for countable exponential-feedback transseries*, AI-assisted research report prepared for Vladimir Reshetnikov, 29 September 2026, ProveIt research package. Conventional, non-Lean results; precise input labels are recorded in Section 7. [Pinned article source](#). Companion scope statement: [Pinned scope statement](#).
- [4] *Negative-Ray Summation of Countable Exponential-Feedback Transseries*, AI-assisted research report prepared for Vladimir Reshetnikov, 29 September 2026, ProveIt research package; README consulted. [Pinned negative-ray README](#).
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- [11] [Editorial addition] *Signed Quadratic-Feedback Inversion: A Finite-Core Resolution and Sharp Entire Borel Growth*, research draft prepared for Vladimir Reshetnikov, ProveIt, filed in batch 49, `Analysis/Transseries/docs/series-and-transseries/Signed_Quadratic_Feedback_Inversion/article.tex`. Claimed proof, not independently reviewed.
- [12] [Editorial addition] *Exact Type Beyond Log-Convexity: A complete reversion criterion, sharp distortion, and dimension-free nonlinear calculus*, research article prepared for Vladimir Reshetnikov, ProveIt, filed in batch 52, `Analysis/Transseries/docs/series-and-transseries/Exact_Weighted_Type_Beyond_Log_Convexity/article.tex`. Not independently reviewed.
- [13] [Editorial addition] *Sharp Subexponential Reversion: Exact extremal cost, a Gevrey transition, and composition of two divergent series*, research article prepared for Vladimir Reshetnikov, ProveIt, filed in batch 52, `Analysis/Transseries/docs/series-and-transseries/Sharp_Subexponential_Cost_Gevrey_Reversion/article.tex`. Not independently reviewed.